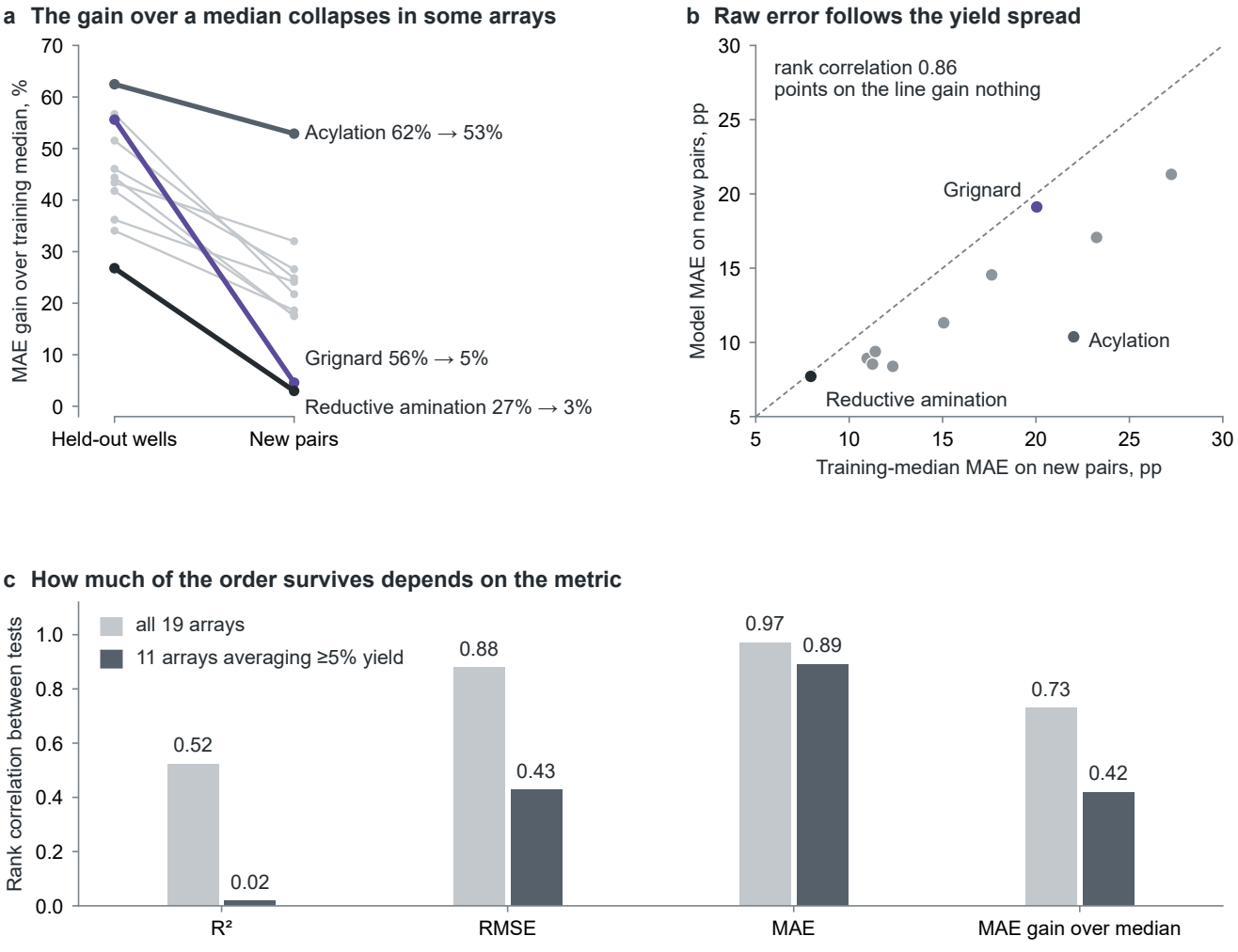
a The gain over a median collapses in some arrays
MAE gain over training median, %
70
60
50
40
30
20
10
0
Acylation 62% → 53%
Grignard 56% → 5%
Reductive amination 27% → 3%
Held-out wells
New pairs
b Raw error follows the yield spread
rank correlation 0.86
points on the line gain nothing
Model MAE on new pairs, pp
30
25
20
15
10
5
Grignard
Acylation
Reductive amination
5
10
15
20
25
30
Training-median MAE on new pairs, pp
c How much of the order survives depends on the metric
Rank correlation between tests
1.0
0.8
0.6
0.4
0.2
0.0
all 19 arrays
11 arrays averaging ≥5% yield
0.52
0.02
0.88
0.43
0.97
0.89
0.73
0.42
R²
RMSE
MAE
MAE gain over median